



LECTURE 6

GROUP TECHNOLOGY

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1. Objectives
2. Review
3. Group technology
4. Parts classification and coding
5. Production flow analysis
6. Conclusion



Objectives

1. Understanding the importance of group technology for efficient production.
2. Presentation of parts classification and coding techniques.
3. Knowledge of the analysis methods for the production flows.



Review

1. Material storage system





Group technology (1)

Group technology is a manufacturing philosophy in which similar parts are identified and grouped together to take advantage of their similarities in design and production.

Similar parts $\Leftrightarrow$ *part families*

Production equipment $\Leftrightarrow$ *machine groups or cells*



Each cell is specialized in production of a part family.

Group technology is most appropriately applied under the following conditions:

- the plant currently uses traditional batch production and a process type layout;
- the parts can be grouped into part families.



Group technology (2)



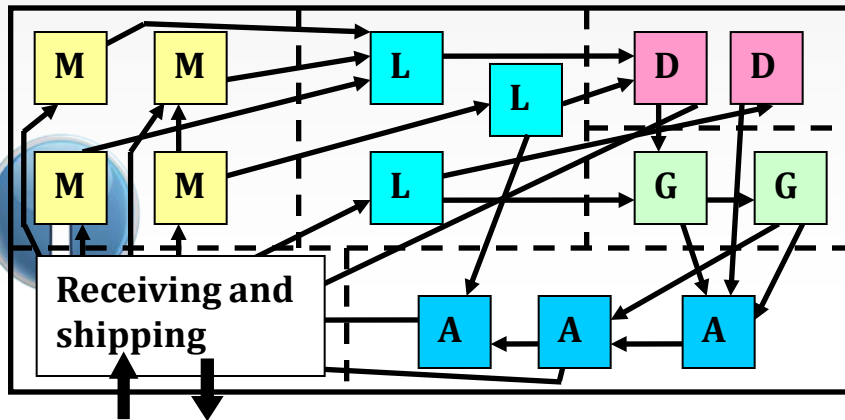
- ❑ Tasks that a company must undertake to implement group technology(GT):
 - **identifying** the part families,
 - **rearranging** production machines into machine cells.

- ❑ The benefits of group technology:
 - GT **promotes standardization** of tooling, fixturing, and setups
 - material **handling is reduced**
 - process **planning** and production **scheduling** are **simplified**
 - **setup times are reduced** => lower manufacturing lead time
 - **work-in-process is reduced**
 - worker **satisfaction** usually **improves**
 - **higher quality** work is accomplished

Part families

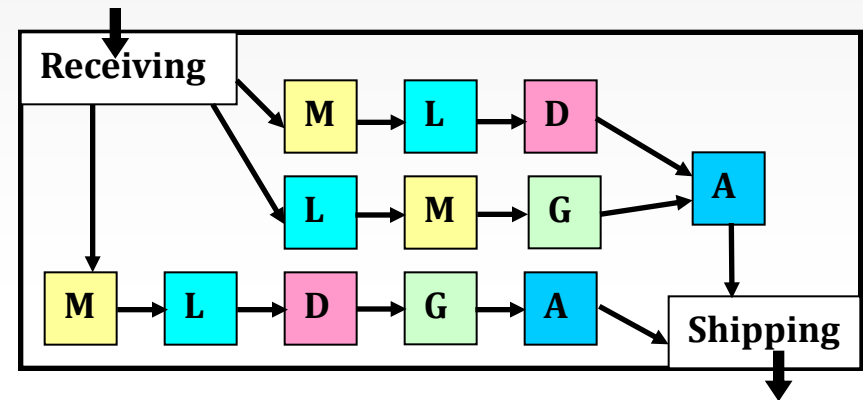
A part family is a collection of parts that are similar either because of geometric shape and size or because similar processing steps are required in their manufacturing.

Process type plant layout



vs

GT layout



Methods used to group the parts into families:

- visual inspection,
- parts classification and coding,
- production flow analysis.



Visual inspection

Visual inspection involves the classification of parts into families by looking at either physical parts or their photographs and arranging them into groups having similar features.



Visual inspection is the least sophisticated and least expensive.





Parts classification and coding (1)

In parts classification and coding method, similarities among parts are identified, and these similarities are related in a coding system.

❑ Parts classification systems:

- system based on part design attributes,
- system based on part manufacturing attributes,
- system based on both design and manufacturing attributes.

Part design attributes

Basic external shape, Basic internal shape, Rotational or rectangular shape, Length-to-diameter ration (rotational parts), Aspect ratio (rectangular parts), Material type, Part function, Major dimensions, Minor dimensions, Tolerances, Surface finish.

Part manufacturing attributes

Major process, Minor process, Operation sequence, Major dimension, Surface finish, Machine tool, Production cycle time, Batch size, Annual production, Fixture required, Cutting tools



Self-assessment test

1. The group technology can not be used for production systems that produce in batches.

T

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2. The group technology applies only if the products can be grouped.

T

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3. The use of group technology decreases the number of operations in the production process.

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4. Visual inspection is the cheapest method used to classify products.

T

F

5. Product classification systems may not use the design features.

T

F

Parts classification and coding (2)

- There are three structures used in classification and coding schemes:
 - **monocode** (or hierarchical structure): - the interpretation of each successive symbol depends on the value of the preceding symbols.
 - **polycode** (or chain-type structure): - the interpretation of each symbol in the sequence is always the same.
 - **hybrid** (or mixed-mode structure): - is a combination between monocode and polycode.

Cylindrical part

Length to diameter ratio

Rectangular part

Length to width ratio



Monocode: **15**

Polycode: **15**



Monocode: **25**

Polycode: **25**

Cylindrical part

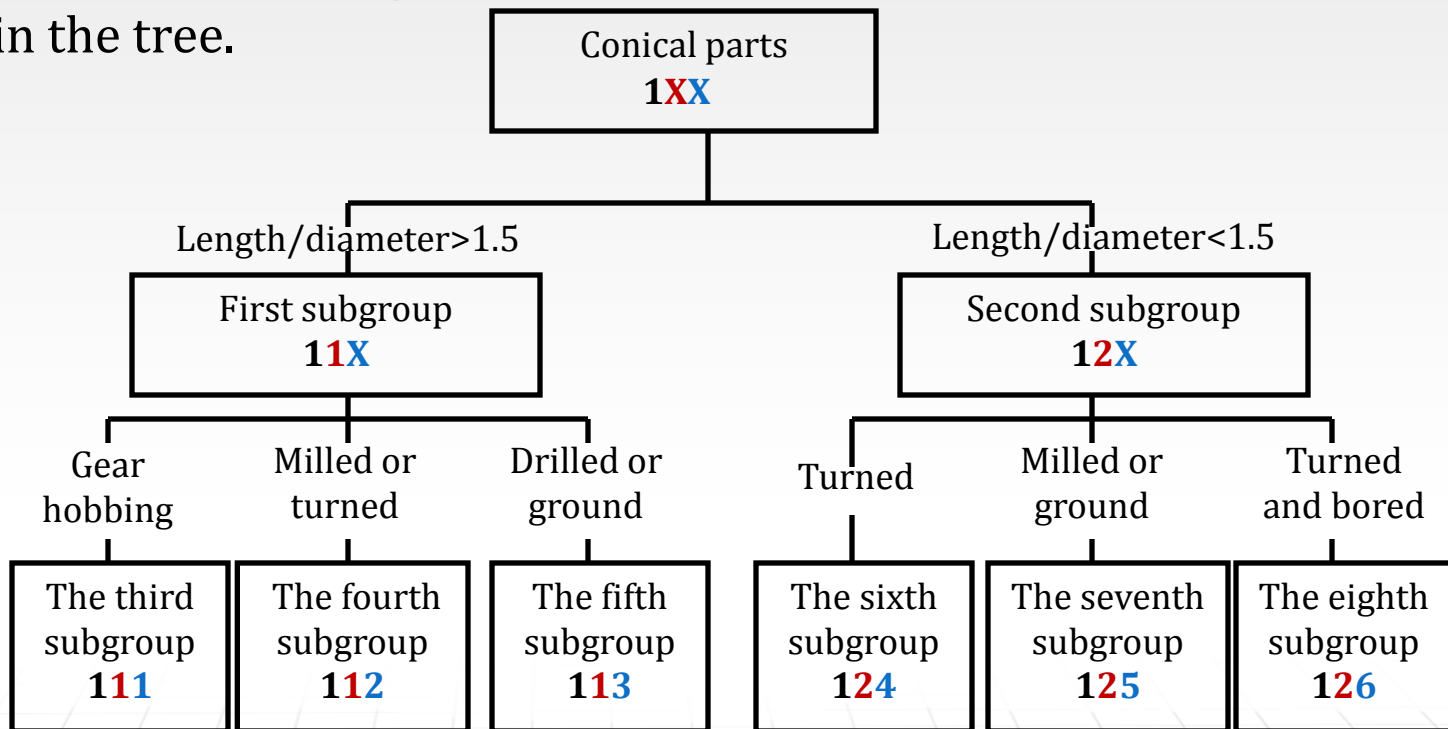
Length of the part

Rectangular part

Length of the part

Monocode

- ❑ The hierarchical code structure (called a monocode), divides all parts of the total population into distinct subgroups of about equal size.
- ❑ The number of digits in the code is determined by the number of levels in the tree.



A few code numbers can represent a large amount of information.

the complexity associated with defining all the branches.

Polycode

- The chain structure, called a polycode, is created from a code table or matrix like the example shown:

Digit	Class of feature	Possible values of digits			
		1	2	3	4
1	External shape	Shape ₁	Shape ₂	Shape ₃	-
2	Internal shape	None	Shape ₁	-	-
3	Number of holes	0	1-2	3-5	6-8
4	Type of holes	Axial	Cross	Axial and cross	
5	Flats	External	Internal	Both	
6	Gear teeth	Spur	Helical		

Example 1

The policode:

3 1 1 4 1 2



Compact and easy to use and develop.



A policode lacks the detail present in a hierarchical structure.



Production flow analysis

Production flow analysis (PFA) is a method for identifying part families and associated machine groupings that uses the information contained on production route sheets rather than on part.



- ❑ PFA uses manufacturing data rather than design data, so two possible anomalies can occur in parts classification and coding:
 - parts whose basic geometries are quite different may nevertheless require similar or even identical process routings.
 - parts whose geometries are quite similar may nevertheless require process routings that are quite different.
- ❑ Steps for PFA:
 - **data collection** (part number, operation sequence, lot size, time standards and annual demand)
 - **sorting of process routings** (parts are arranged into groups)
 - **PFA chart** (or part-machine incidence matrix)
 - **cluster analysis** (related groupings are identified and rearranged)

Grouping parts and machines by rank order clustering

Machines	Parts								
	1	2	3	4	5	6	7	8	9
1	1	0	0	1	0	0	0	1	0
2	0	0	0	0	1	0	0	0	1
3	0	0	1	0	1	0	0	0	1
4	0	1	0	0	0	1	0	0	0
5	1	0	0	0	0	0	0	1	0
6	0	0	1	0	0	0	0	0	1
7	0	1	0	0	0	1	1	0	0

PFA require less time than a complete parts classification and coding procedure.

Data used are derived from existing production route sheets.

Rank order clustering algorithm

STEP 1:

- Read the series of 1's and 0's in each row of the matrix from left to right as a binary number and then rank the rows in order of decreasing value. In case of a tie, rank the rows in the same order as they are in the current matrix;
- Number the rows from top to bottom. Is the current order of rows the same as the rank order determined in previous step? If YES STOP, else go to the next step;
- Reorder the rows in the part-machine incidence matrix by listing them in decreasing rank order, starting from the top.

Binary values	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰		
	Parts									Value	Rank
Machines	1	2	3	4	5	6	7	8	9		
1	1	0	0	1	0	0	0	1	0	290	1
2	0	0	0	0	1	0	0	0	1	17	7
3	0	0	1	0	1	0	0	0	1	81	5
4	0	1	0	0	0	1	0	0	0	136	4
5	1	0	0	0	0	0	0	1	0	258	2
6	0	0	1	0	0	0	0	0	1	65	6
7	0	1	0	0	0	1	1	0	0	140	3



Binary values	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰		Rank
	Parts										
Machines	1	2	3	4	5	6	7	8	9		
1	1	0	0	1	0	0	0	1	0	1	
5	1	0	0	0	0	0	0	1	0	2	
7	0	1	0	0	0	1	1	0	0	3	
4	0	1	0	0	0	1	0	0	0	4	
3	0	0	1	0	1	0	0	0	1	5	
6	0	0	1	0	0	0	0	0	1	6	
2	0	0	0	0	1	0	0	0	1	7	

Rank order clustering algorithm

STEPL 2:

- Read the series of 1's and 0's in each column of the matrix from top to bottom as a binary number. Rank the columns in order of decreasing value.
- Number the columns from left to right. Is the current order of columns the same as the rank order determined in the previous step? If YES, STOP, else go to next step.
- Reorder the columns in the part-machine incidence matrix by listing them in decreasing rank order, starting with the left column. Go to beginning.

Machines	Parts									Binary values
	1	2	3	4	5	6	7	8	9	
1	1	0	0	1	0	0	0	1	0	2^6
5	1	0	0	0	0	0	0	1	0	2^5
7	0	1	0	0	0	1	1	0	0	2^4
4	0	1	0	0	0	1	0	0	0	2^3
3	0	0	1	0	1	0	0	0	1	2^2
6	0	0	1	0	0	0	0	0	1	2^1
2	0	0	0	0	1	0	0	0	1	2^0
Value	96	24	6	64	5	24	16	96	7	
Rank	1	4	8	3	9	5	6	2	7	



Machines	Parts									Binary values
	1	8	4	2	6	7	9	3	5	
1	1	1	1	0	0	0	0	0	0	2^6
5	1	1	0	0	0	0	0	0	0	2^5
7	0	0	0	1	1	1	0	0	0	2^4
4	0	0	0	1	1	0	0	0	0	2^3
3	0	0	0	0	0	0	1	1	1	2^2
6	0	0	0	0	0	0	1	1	0	2^1
2	0	0	0	0	0	0	1	0	1	2^0
Rank	1	2	3	4	5	6	7	8	9	

Rank order clustering algorithm

STEP 3:

- Repeat steps 1 and 2 until there are no more changes in ranking from one iteration to another.
- Group the machines so that the groups form a diagonal.

Binary values	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰		
Machines	Parts									Value	Rank
	1	8	4	2	6	7	9	3	5		
1	1	1	1	0	0	0	0	0	0	448	1
5	1	1	0	0	0	0	0	0	0	384	2
7	0	0	0	1	1	1	0	0	0	56	3
4	0	0	0	1	1	0	0	0	0	48	4
3	0	0	0	0	0	0	1	1	1	7	5
6	0	0	0	0	0	0	1	1	0	6	6
2	0	0	0	0	0	0	1	0	1	5	7



Binary values	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰		
Machines	Parts									Rank	
	1	8	4	2	6	7	9	3	5		
1	1	1	1	0	0	0	0	0	0	1	
5	1	1	0	0	0	0	0	0	0	2	
7	0	0	0	1	1	1	0	0	0	3	
4	0	0	0	1	1	0	0	0	0	4	
3	0	0	0	0	0	0	1	1	1	5	
6	0	0	0	0	0	0	1	1	0	6	
2	0	0	0	0	0	0	1	0	1	7	

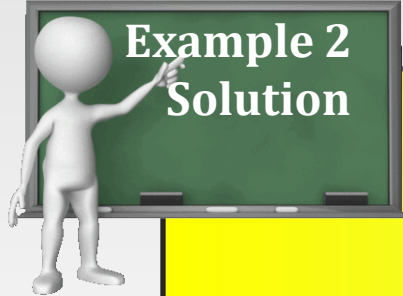
Grouping parts and machines by rank order clustering

Example 2

Consider the part-machine incidence matrix shown below. This is the same as the original part-machine matrix (shown in previous example) except that part number 2 requires processing on machines 1, 4 and 7, and part number 4 now requires processing on machines 1 and 4. Use the rank order clustering technique to arrange parts and machine into groups.

Machines	Parts								
	1	2	3	4	5	6	7	8	9
1	1	1	0	1	0	0	0	1	0
2	0	0	0	0	1	0	0	0	1
3	0	0	1	0	1	0	0	0	1
4	0	1	0	1	0	1	0	0	0
5	1	0	0	0	0	0	0	1	0
6	0	0	1	0	0	0	0	0	1
7	0	1	0	0	0	1	1	0	0

Grouping parts and machines by rank order clustering



The rank order clustering technique converges to a solution in two iterations:

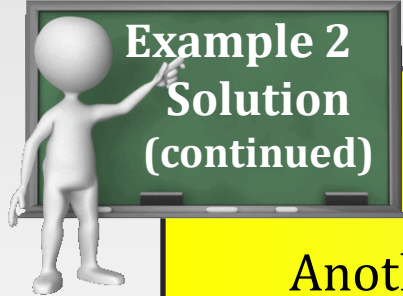
The elements “21” and “41” (part 2 processed on machine 1, part 4 processed on machine 1) are exceptional elements..

Binary values	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	Rank
Machines	Parts									
	1	8	2	4	6	7	9	3	5	
1	1	1	1	1	0	0	0	0	0	1
5	1	1	0	0	0	0	0	0	0	2
4	0	0	1	1	1	0	0	0	0	3
7	0	0	1	0	1	1	0	0	0	4
3	0	0	0	0	0	0	1	1	1	5
6	0	0	0	0	0	0	1	1	0	6
2	0	0	0	0	0	0	1	0	1	7

¹King recommends to replace them by an asterisk and treat them as zeros when applying the rank order clustering technique.

¹KING, J.R., “Machine-Component Grouping in Production Flow Analysis ; an Approach Using a Rank Order Clustering Algorithm”, International Journal of production Research, Vol. 18, 1980, pp 213-222

Grouping parts and machines by rank order clustering



Example 2 Solution (continued)

The solution offered by King is not effective!

Another way of dealing with the overlap is to duplicate the machine that is used by more than one family (machine 1).

Binary values	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	Rank
Machines	Parts									
	1	8	2	4	6	7	9	3	5	
1a	1	1	0	0	0	0	0	0	0	1
5	1	1	0	0	0	0	0	0	0	2
4	0	0	1	1	1	0	0	0	0	3
1b	0	0	1	1	0	0	0	0	0	
7	0	0	1	0	1	1	0	0	0	4
3	0	0	0	0	0	0	1	1	1	5
6	0	0	0	0	0	0	1	1	0	6
2	0	0	0	0	0	0	1	0	1	7



Arranging machines in a GT cell

- ❑ After part-machine groupings have been identified the problem is to organize the machines into the most logical arrangement.
- ❑ Hollier introduces six heuristic approaches to solving the machine arrangement problem.
- ❑ First two methods will be described using data contained in From-To charts.





Hollier method 1

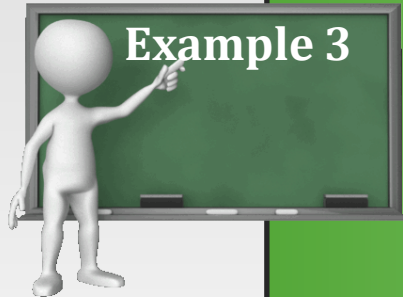
The method requires next steps:

1. Develop the From-To chart from part routing data;
2. Determine the “From” and “To” sums for each machine;
3. Assign machines to the cell based on minimum “From” or “To” sums:
 - if the minimum value is a “To” sum, then the machine is placed at the beginning of the sequence;
 - if the minimum value is a “From” sum then the machine is placed at the end of the sequence;

Tie breaker rules:

- if a tie occurs between minimum “To” sums or minimum “From” sums => the machine with the minimum “From/To” ratio is selected;
 - if both “To” and “From” sums are equal for a selected machine, it is passed over and the machine with the next lowest sum is selected;
 - if a minimum “To” sum is equal to a minimum “From” sum, then both machines are selected and placed at the beginning and end of the sequence, respectively.
4. Reformat the From-To chart :
 - after each machine has been selected, restructure the From-To chart by eliminating the row and column corresponding to the selected machine and recalculate the “From” and “To” sums;
 - repeat steps 3 and 4 until all machines have been assigned.

Hollier method 1

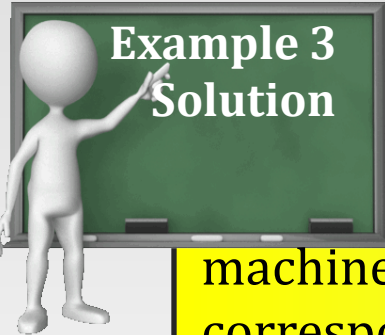


Example 3

Suppose that four machines 1,2,3 and 4 have been identified as belonging in a GT machine cell. An analysis of 50 parts processed on these machines has been summarized in the “From-To” chart below. Additional information is that 50 parts enter the machine grouping at machine 3, 20 parts leave processing at machine 1, and 30 parts leave after machine 4. Determine a logical machine arrangement using Hollier method 1.

To From	1	2	3	4
1	0	5	0	25
2	30	0	0	15
3	10	40	0	0
4	10	0	0	0

Hollier method 1



“From” and “To” sums after first iteration are illustrated in Table 1. => The minimum sum value is the “To” sum for machine 3. => Machine 3 is placed at the beginning of sequence and the corresponding row and column are eliminated resulting the Table 2

To From	1	2	3	4	Sums “from”
1	0	5	0	25	30
2	30	0	0	15	45
3	10	40	0	0	50
4	10	0	0	0	10
Sums “To”	50	45	0	40	

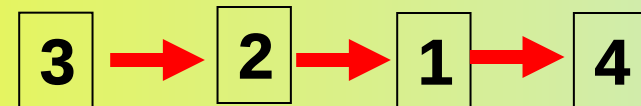
Table 1. From and To sums; first iteration

To From	1	2	4	Sums “from”
1	0	5	25	30
2	30	0	15	45
4	10	0	0	10
Sums “To”	40	5	40	

Table 2. From and To sums; second iteration

To From	1	4	Sums “From”
1	0	25	25
4	10	0	10
Sums “To”	10	25	

Table 3. From and To sums; third iteration





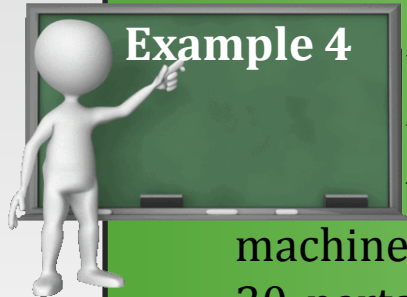
Hollier method 2



The method can be reduced to three steps:

1. Develop the From-To chart
 2. Determine the From/To ratio for each machine:
 - Sum all “From” trips and “To” trips for each machine. The “From” sum for a machine is determined by adding the entries in the corresponding row, and the “To” sum by adding the entries in the corresponding column.
 - Calculate From/To ratio for each machine
 3. Arrange machines in order of decreasing From/To ratio:
 - In case of a tie, the machine with the higher “From” value is placed ahead of the machine with a lower value.
- 

Hollier method 2



Example 4

Suppose that four machines 1,2,3 and 4 have been identified as belonging in a GT machine cell. An analysis of 50 parts processed on these machines has been summarized in the “From-To” chart of table below. Additional information is that 50 parts enter the machine grouping at machine 3, 20 parts leave processing at machine 1, and 30 parts leave after machine 4. Determine a logical machine arrangement using Hollier method 2.

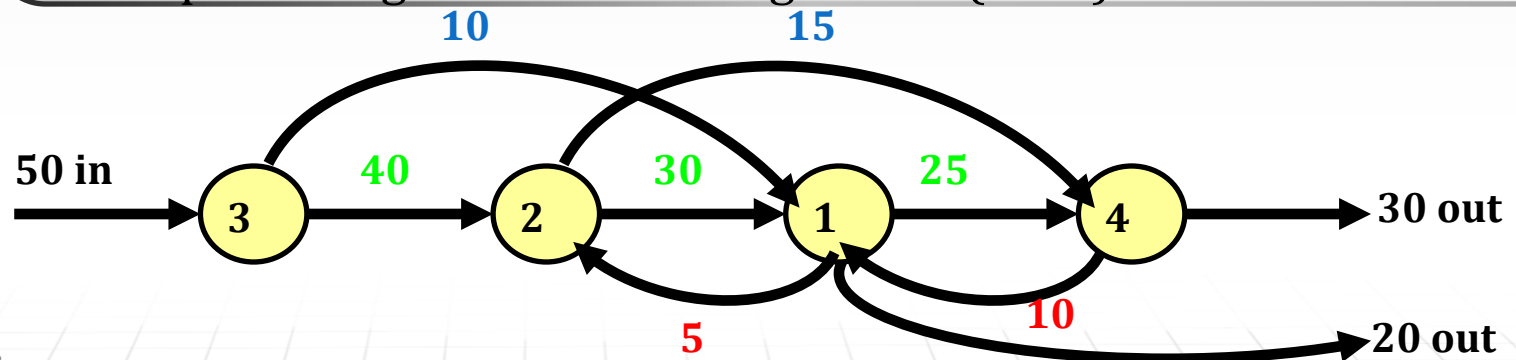
To From	1	2	3	4
1	0	5	0	25
2	30	0	0	15
3	10	40	0	0
4	10	0	0	0

To From	1	2	3	4	„From” sums	“From”/”To”
1	0	5	0	25	30	0.6
2	30	0	0	15	45	1.0
3	10	40	0	0	50	∞
4	10	0	0	0	10	0.25
„To” sums	50	45	0	40	135	

3 → 2 → 1 → 4

The work flow in the cell

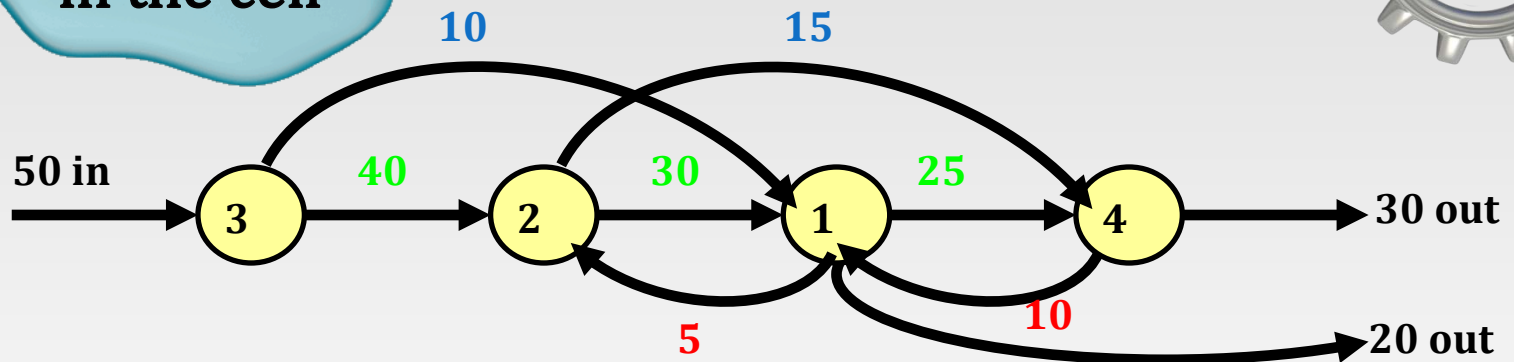
- ❑ It is helpful to use a graphical techniques, such as the flow diagram to conceptualize the work flow in the cell.
- ❑ The flow diagram for the machine arrangement in examples 3 and 4 is illustrated below:
- ❑ Two performance measures can be defined to compare solutions to the machine sequencing problem:
 - percentage of in-sequence moves (PIM)
 - percentage of backtracking moves (PBM)



$$\text{PIM} = \frac{\sum (\text{in sequence moves})}{\text{total number of moves}} \times 100\%$$

$$\text{PBM} = \frac{\sum (\text{backtracking moves})}{\text{total number of moves}} \times 100\%$$

The workflow
in the cell



$$\text{PIM} = \frac{\sum(\text{in sequence moves})}{\text{total number of moves}} \times 100\%$$

$$\text{PBM} = \frac{\sum(\text{backtracking moves})}{\text{total number of moves}} \times 100\%$$

$$\text{PIM} = \frac{40 + 30 + 25}{40 + 30 + 25 + 10 + 15 + 10 + 5} \times 100\% = \frac{95}{135} \times 100\% = 70.37\%$$

$$\text{PBM} = \frac{10 + 5}{40 + 30 + 25 + 10 + 15 + 10 + 5} \times 100\% = \frac{15}{135} \times 100\% = 11.11\%$$



Self-assessment test

1. In the case of polycode the interpretation of each symbol from the sequence depends on the previous symbol.

T

F

2. For the chain-type structure the interpretation of each symbol in the sequence is always the same.

T

F

3. For the polycode structure, the number of digits of the code is determined by the number of levels in the hierarchy.

T

F

4. Monocode encodes more information.

T

F

5. Sorting of process routings is a step in the production flow analysis.

T

F



Conclusion

- ☐ Using group technology provides a more efficient use of resources.
- ☐ Product group activity is more difficult as increasing the number of product types.
- ☐ Production flow analysis is necessary to streamline production activities.



- ☐ The components of a Flexible Manufacturing System (FMS)
- ☐ The applications and benefits of FMS
- ☐ FMS planning and design
- ☐ Quantitative analysis of FMS